

LAPD Crime Trends: Analyzing Temporal Patterns and Environmental Influences



Jeopardy Officer Lane



**Vice Investigator
Bourgeois**



**Watch Commander
Bocharova**

Objectives

- Leverage LAPD OpenData to conduct a comprehensive analysis of crime reports across Los Angeles.
- Examine trends in crime based on demographic and geographic variables
- Train a machine learning model to analyze the data and develop linear regression models that can understand crime reporting dynamics



Disclaimer:

This is a weighty and serious subject that may diminish any hard-earned optimism.

Our use of animal imagery and whimsical slide titles is an effort to lighten the subject matter, preventing the project from becoming overwhelmingly somber.



LAPD OpenData 2020-2024

Map:
Los Angeles - Top 20 Crimes
Reported 2020-2024

- Established in 2010
- Redesigned in 2024 to comply with FBI NIBRS reporting requirements
- NIBRS
 - captures details on each single crime incident—as well as on separate offenses within the same incident
 - information on victims, known offenders, relationships between victims and offenders, arrestees, and
 - property involved in crimes.

Crime Map 2020-2024



Data Cleaning

Dataset Overview

- 1,005,104 Individual crime records
- 28 columns
- 141 Crime codes

Data Cleaning & Categorization:

- Normalized key columns (dates, victim details)
- Addressed missing/inconsistent values
- Decoded Part 1 crimes; classified Part 2 crimes
- Mapped crime subcategories for deeper pattern analysis

Three major categories:

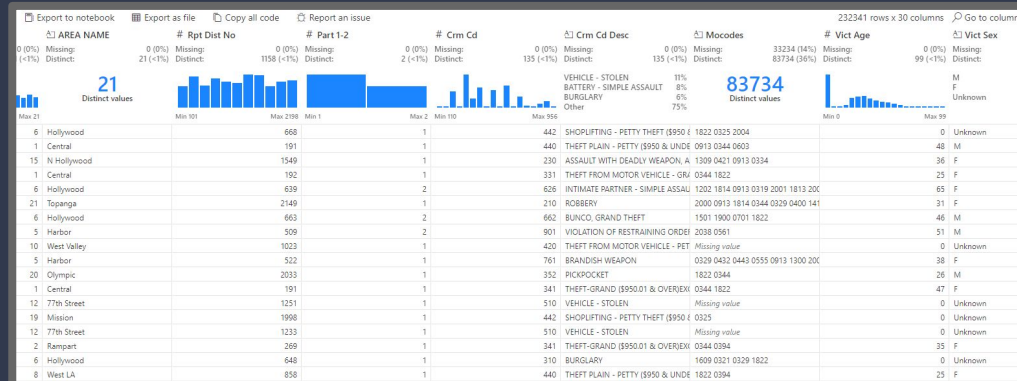
- Violent Crimes,
- Property Crimes,
- Crimes Against Society

Mapped Crime Subcategories:

- Homicide
- Sex Offenses
- Physical & Psychological Violence
- Property Crimes & Fraud
- Illegal Human Abduction & Trafficking
- Government & Public Order Violations



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Correcting Time Format

```
import re
#creating copy of our DF
crime_df_edt = crime_df.copy()
# Convert 'DATE OCC' to datetime format
crime_df_edt['DATE OCC'] = pd.to_datetime(crime_df_edt['DATE OCC'])

# removing all single quotes with Regex from "TIME OCC" column
# "[^d]" - match any character that is NOT a digit
crime_df_edt['TIME OCC'] = crime_df_edt['TIME OCC'].astype(str).replace(r"[^d]", "", regex=True)
# Ensure 'TIME OCC' is a 4-digit string (e.g., '930' -> '0930')
crime_df_edt['TIME OCC'] = crime_df_edt['TIME OCC'].astype(str).zfill(4)

# Convert 'TIME OCC' into HH:MM:SS format
crime_df_edt['time_formatted'] = crime_df_edt['TIME OCC'].str[:2] + ":" + crime_df_edt['TIME OCC'].str[2:] + ":00"
# Combine 'DATE OCC' and formatted time into a single timestamp
crime_df_edt['crime_timestamp'] = pd.to_datetime(crime_df_edt['DATE OCC']).astype(str) + " " + crime_df_edt['time_formatted']

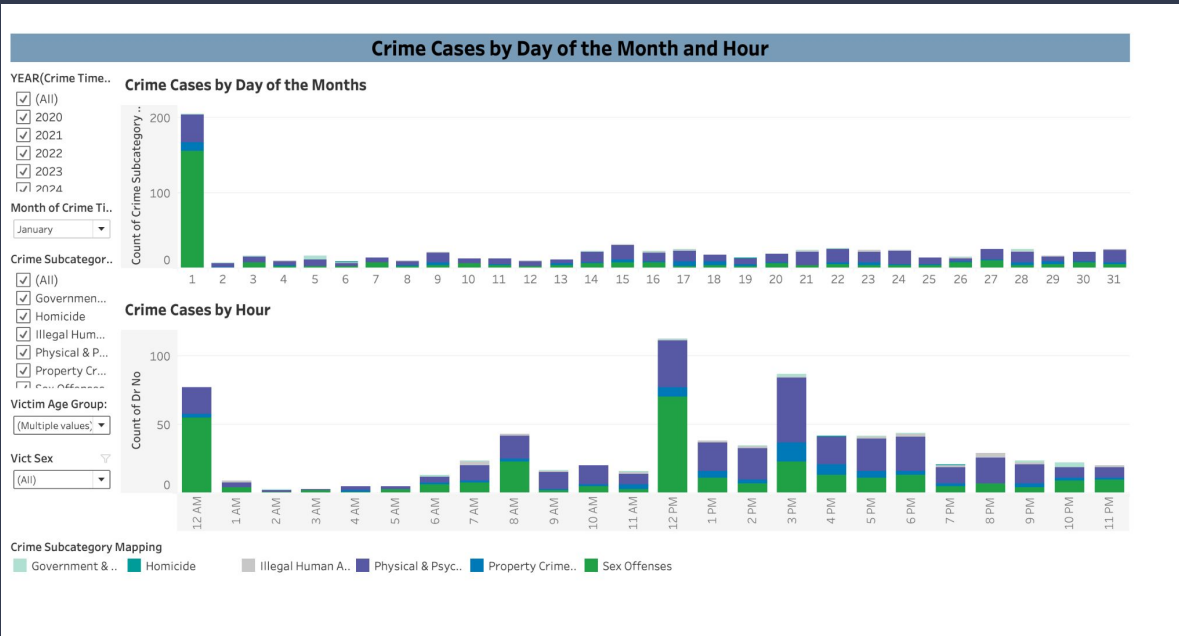
# Drop the intermediate column 'time_formatted' (optional)
crime_df_edt.drop(columns=['time_formatted'], inplace=True)

crime_df_edt.head()
```

Python

... /var/folders/cf/117zp0s7vdbnznx_vs2kls8000gn/T/ipykernel_15678/3774130617.py:5: UserWarning: Could not infer format, so each

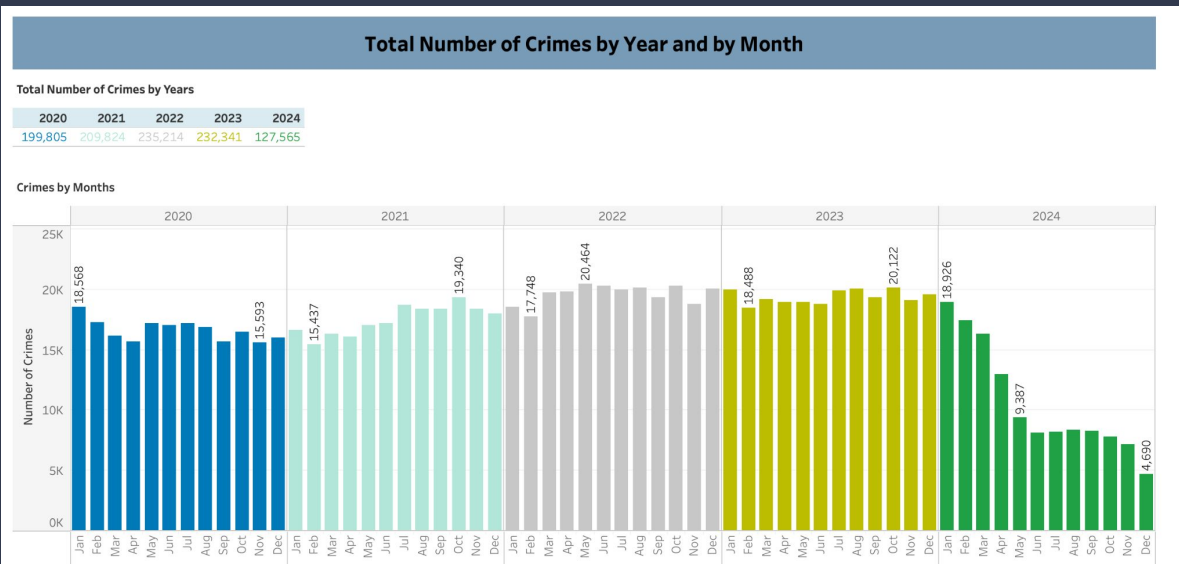
Patterns and Trends Insights



← **Statistics:** January. Minors.

Other insights:

- Crime rates show an increase on the 1st day of the month (especially January)
- Pattern emerges around seasonal solar events (Spring and Fall Equinox and Summer and Winter Solstices), where crime rates frequently spike on or immediately after these dates.
- Homicides are most frequently happens between 9 PM and 2 AM. With another peak at 2PM. The 4th day of the month showed the highest number of homicides over the five years studied. And Weekends has highest rate of it among whole week.



← Statistics: by Months and Years



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Trends

Holidays - Lunar Phases - Wildfires*

Holidays: reporting spikes around, but not necessarily on the holiday

Lunar Phases: myth debunked

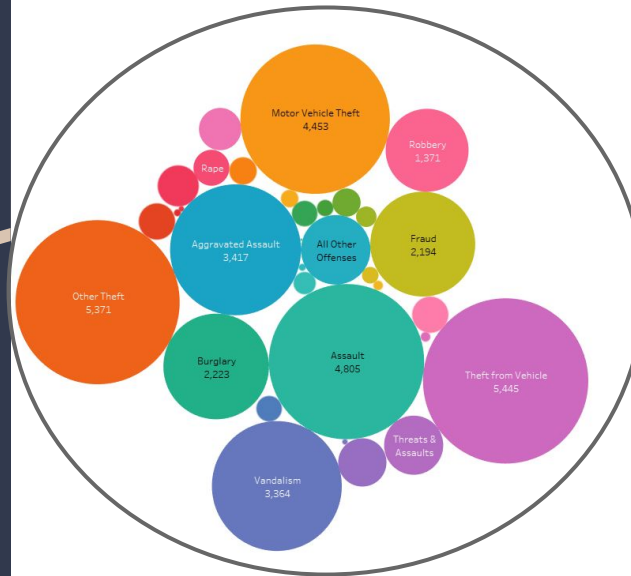
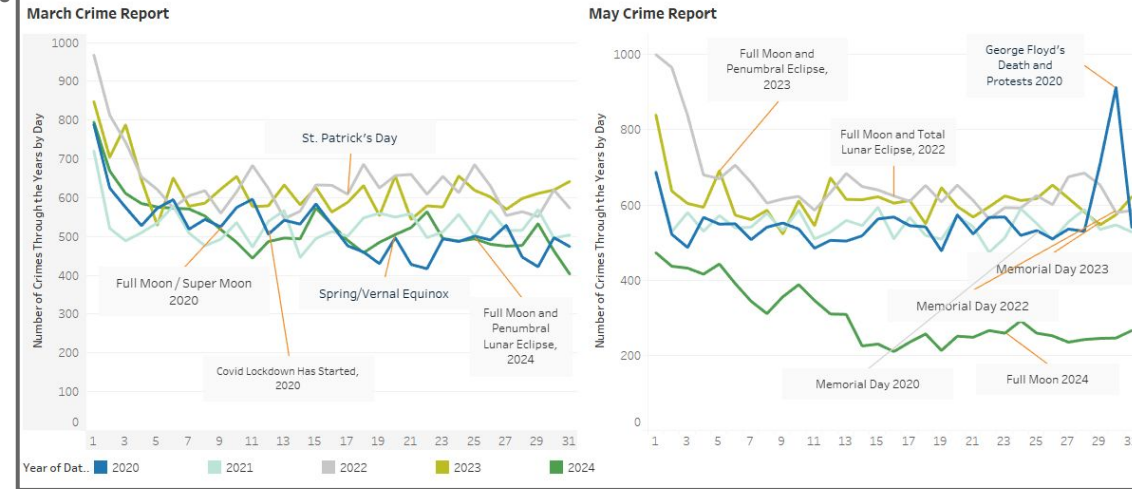
Wildfires: property crime increases



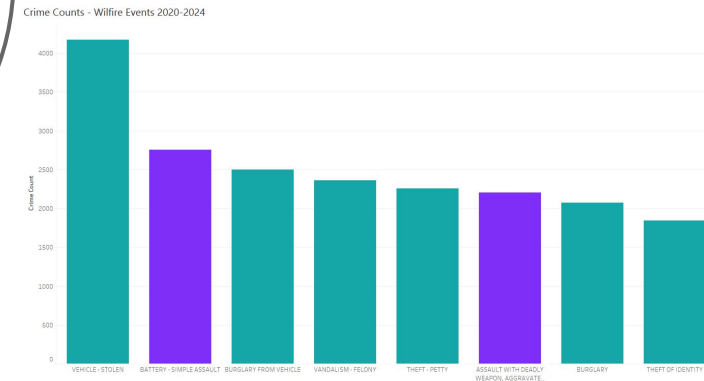
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*wildfire-specific data covers the two weeks following the fire's onset

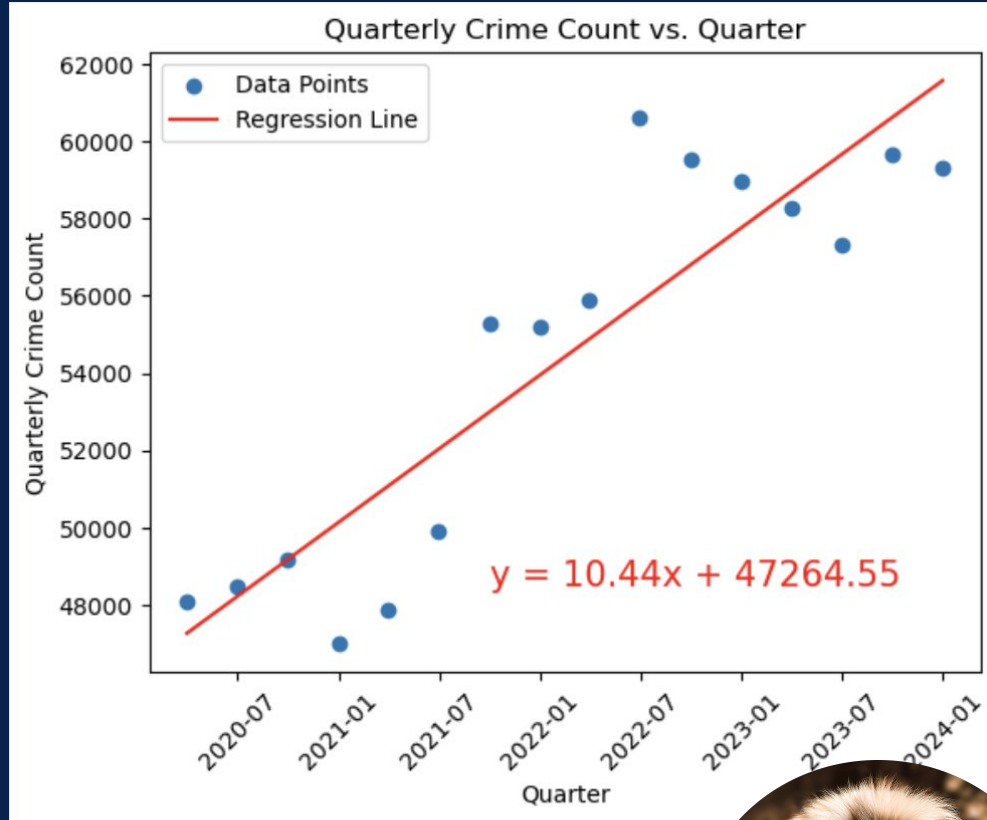
Monthly Reports - Holidays & Lunar Phases



**Wildfire Events – Most
Prevalent Crimes Reported**



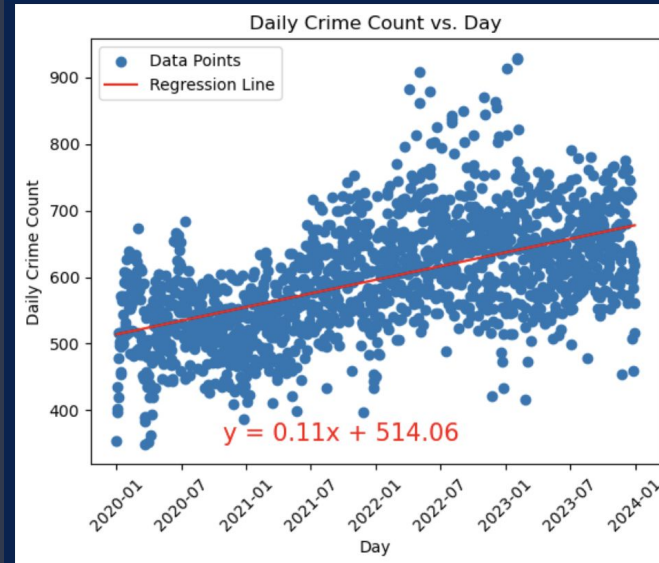
Data Modeling



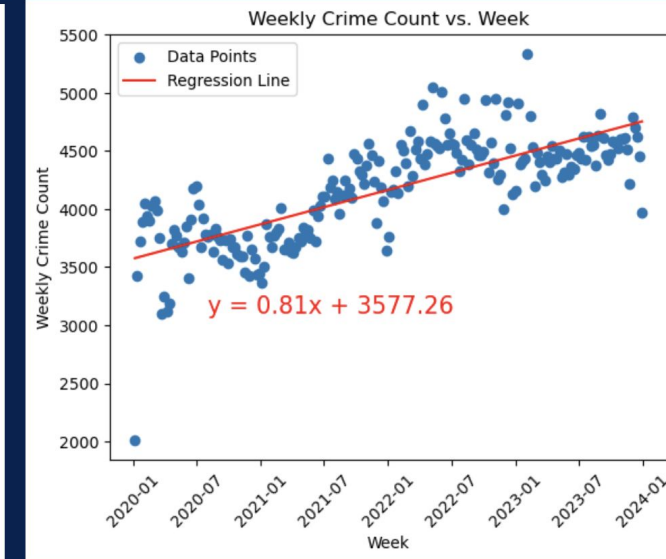
The r-squared is: 0.8014064711287323



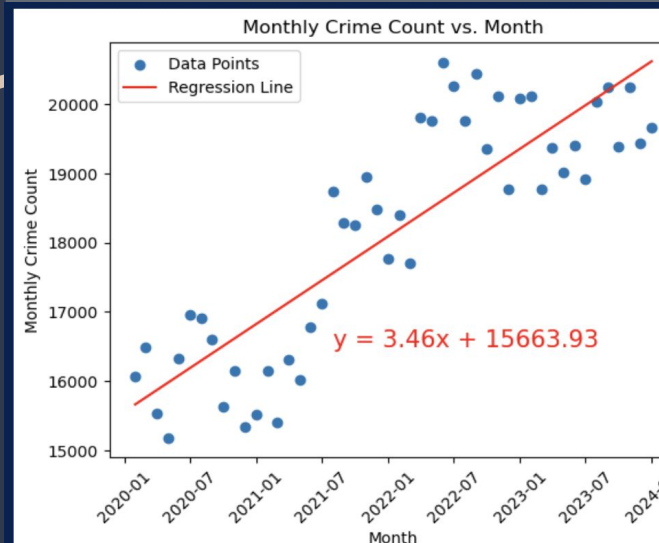
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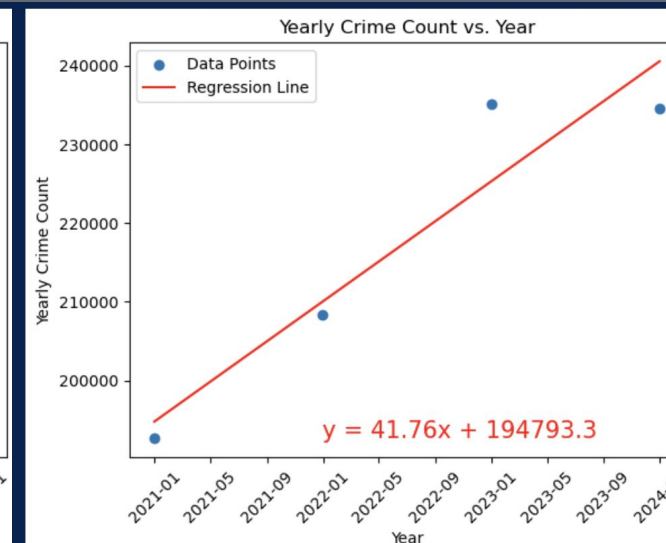
The r-squared is: 0.3013356237271663



The r-squared is: 0.5788230261180906



The r-squared is: 0.7309737052287066



The r-squared is: 0.8938341862508675

Data Modeling Disasters or Rage Against the Machine (Learning)



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5748/5748 – 1s – loss: 0.1511 – accuracy: 0.9649 – 1s/epoch – 202us/step
Loss: 0.1511373072862625, Accuracy: 0.9649178385734558

```
from sklearn.metrics import classification_report

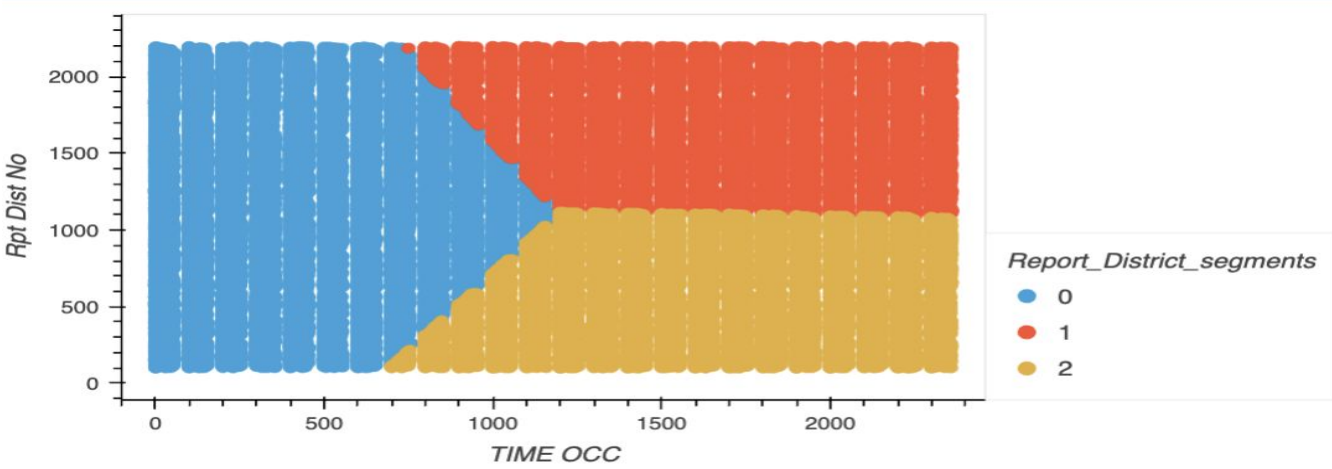
y_pred = nn.predict(X_test)
y_pred_classes = (y_pred > 0.5).astype(int)

print(classification_report(y_test, y_pred_classes))
```

✓ 1.7s

5748/5748 [=====] – 1s 195us/step

	precision	recall	f1-score	support
0	0.96	1.00	0.98	177459
1	0.00	0.00	0.00	6452
accuracy			0.96	183911
macro avg	0.48	0.50	0.49	183911
weighted avg	0.93	0.96	0.95	183911



Conclusions & Next Steps



Conclusion:

Crime is dynamic and people are unpredictable.

Next Steps:

Further research incorporating socio-political data, economic indicators, and psychological factors could yield deeper insights into the underlying influences on crime patterns.

Citations:

Our project dataset:

[Crime Data from 2020 to Present | Los Angeles](#)

Additional Resources:

<https://lasd.org/transparency/part1and2crimedata/Griffithobservatory.org>

<https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr>

THE SHOUT OUT TO ANDREW SLIDE



Ladies and Gentlemen,

Today, we gather to honor the exceptional service and dedication of Jeopardy Office Andrew Lane. His unwavering perseverance and commitment to our machine learning model is truly commendable. Each day, Jeopardy Office Lane faces challenges with courage, professionalism, and resilience, ensuring our sanity the minimum accuracy requirement for this project.

On behalf of Group 4, we extend our deepest gratitude and appreciation for your tireless efforts. Your service is not just a job, but a calling, and we are privileged to have such a dedicated individual in our ranks.

In honor of this, please accept this promotion to Mounted Service Officer.

Thank you for your service, Mounted Service Officer Lane, and congratulations on your well-deserved commendation.



QUESTIONS?